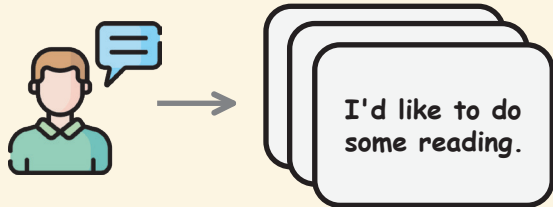


Inputs

Everyday Human Demands



Current 4-View Obs.



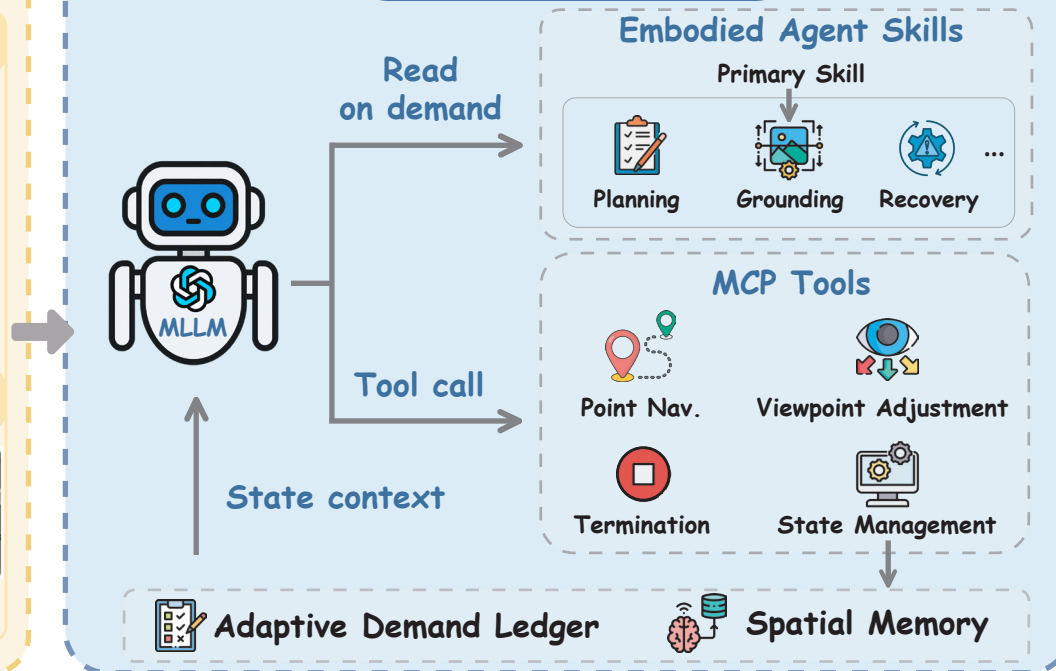
FRONT

RIGHT

BACK

LEFT

DemandAgent



Outputs

Target point selection



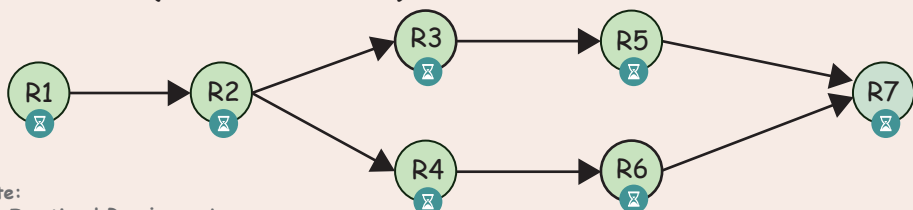
Nav. tool

Target reached



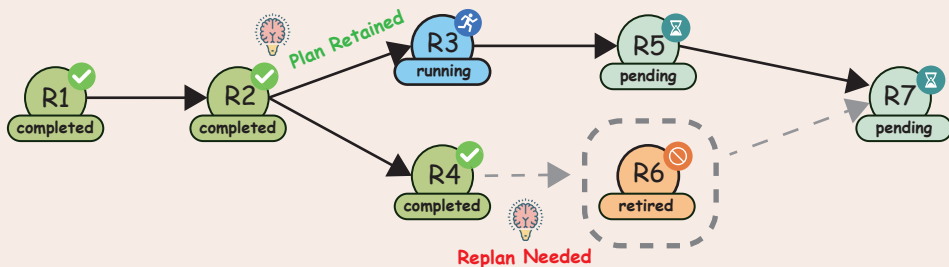
Adaptive Demand Ledger

Initialization (before interaction)



Note:
R = Functional Requirement

Update (during interaction)



Spatial Memory



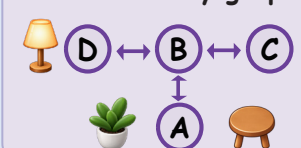
Unexplored Areas

Recall results

- 1) Lamp seen in Area D
- 2) Route cue: Area A ↔ Area B ↔ Area D

Where did I see the lamp?

Online memory graph



Graph Node A

Area: A (new)
Landmark: plant, stool
Type: balcony (inferred)

Graph Edge AB

Doorway: A ↔ B
Appearance: brown
Traversal state: ✓